

Decision Tree Induction: Attribute Selection Measure: Information Gain (ID3/C4.5)

age	income	student	credit rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

✓ Step 1: Dataset Summary

Target: buys_computer (yes/no)

Total records = 14

Count:

- Yes = 9
- No = 5

✓ Step 2: Entropy of Dataset

$$\begin{aligned} \text{Entropy}(S) &= -p_+ \log_2(p_+) - p_- \log_2(p_-) \\ &= -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) \\ &= -(0.643)(-0.64) - (0.357)(-1.48) \\ &= 0.940 \end{aligned}$$

✓ Step 3: Compute Information Gain for Each Attribute

◆ (A) Attribute: AGE

Values:

- ≤ 30 (5 records: 2 yes, 3 no)
 - 31–40 (4 records: 4 yes, 0 no)
 - 40 (5 records: 3 yes, 2 no)
-

Entropy calculations:

AGE ≤ 30 :

$$Entropy = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.971$$

AGE 31–40:

Pure (all yes):

$$Entropy = 0$$

AGE > 40 :

$$Entropy = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.971$$

Weighted Entropy:

$$E(AGE) = \frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) = 0.693$$

Information Gain:

$$Gain(AGE) = 0.940 - 0.693 = 0.247$$

◆ (B) Attribute: INCOME

Values:

- High (4: 2 yes, 2 no)
 - Medium (6: 4 yes, 2 no)
 - Low (4: 3 yes, 1 no)
-

Entropies:

High:

$$Entropy = 1.0$$

Medium:

$$Entropy = 0.918$$

Low:

$$Entropy = 0.811$$

Weighted Entropy:

$$E(INCOME) = \frac{4}{14}(1) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811) = 0.911$$

Gain:

$$Gain(INCOME) = 0.940 - 0.911 = 0.029$$

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◆ (C) Attribute: STUDENT

Values:

- Yes (7: 6 yes, 1 no)
 - No (7: 3 yes, 4 no)
-

Entropies:

Student = Yes:

$$Entropy = 0.592$$

Student = No:

$$Entropy = 0.985$$

Weighted Entropy:

$$E(STUDENT) = \frac{7}{14}(0.592) + \frac{7}{14}(0.985) = 0.789$$

Gain:

$$Gain(STUDENT) = 0.940 - 0.789 = 0.151$$

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◆ (D) Attribute: CREDIT RATING

Values:

- Fair (8: 6 yes, 2 no)
- Excellent (6: 3 yes, 3 no)

Entropies:

Fair:

$$Entropy = 0.811$$

Excellent:

$$Entropy = 1.0$$

Weighted Entropy:

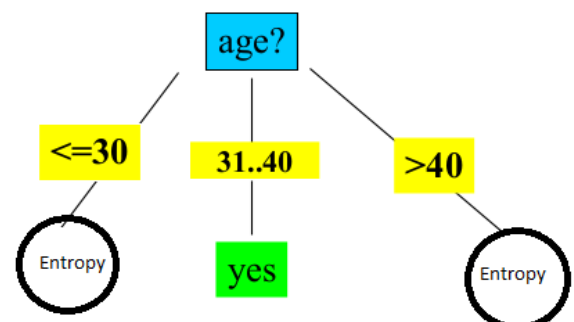
$$E(CREDIT) = \frac{8}{14}(0.811) + \frac{6}{14}(1) = 0.892$$

Gain:

$$Gain(CREDIT) = 0.940 - 0.892 = 0.048$$

✅ Step 4: Choose Root Node

Attribute	Gain
AGE	0.247 (highest)
STUDENT	0.151
CREDIT	0.048
INCOME	0.029



Node: AGE ≤ 30

AGE	INCOME	STUDENT	CREDIT RATING	BUY_COMPUTER
≤30	high	no	fair	no
≤30	high	no	excellent	no
≤30	medium	no	fair	no
≤30	low	yes	fair	yes
≤30	medium	yes	excellent	yes

Count:

- Yes = 2
- No = 3

$$Entropy(S) = 0.971$$

◆ Try splitting on STUDENT

Split:

STUDENT = YES

low, yes, fair → yes
medium, yes, excellent → yes

- Yes = 2, No = 0

$$Entropy = 0$$

STUDENT = NO

high, no, fair → no
high, no, excellent → no
medium, no, fair → no

- Yes = 0, No = 3

$$Entropy = 0$$

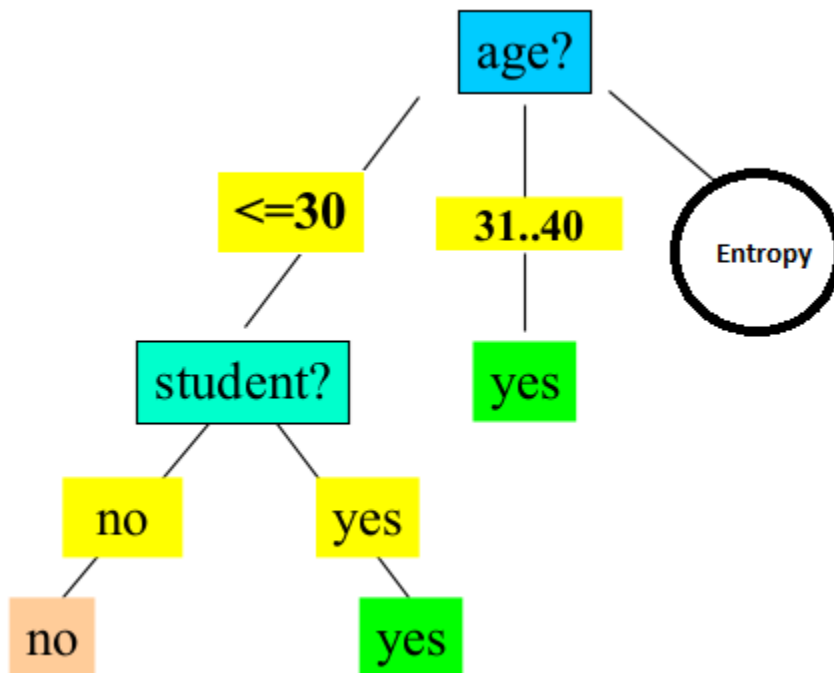
Decision Tree Induction: Attribute Selection Measure: Information Gain (ID3/C4.5)

◆ Weighted Entropy:

$$E = \frac{2}{5}(0) + \frac{3}{5}(0) = 0$$

◆ Information Gain:

$$Gain = 0.971 - 0 = 0.971$$



Node: AGE>41

AGE	INCOME	STUDENT	CREDIT RATING	BUY_COMPUTER
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
>40	medium	yes	fair	yes
>40	medium	no	excellent	no

Decision Tree Induction: Attribute Selection Measure: Information Gain (ID3/C4.5)

Count:

- Yes = 3
- No = 2

$$Entropy = 0.971$$

◆ Try splitting on CREDIT RATING

CREDIT = FAIR

medium, no, fair → yes
low, yes, fair → yes
medium, yes, fair → yes

- Yes = 3, No = 0

$$Entropy = 0$$

CREDIT = EXCELLENT

low, yes, excellent → no
medium, no, excellent → no

- Yes = 0, No = 2

$$Entropy = 0$$

◆ Weighted Entropy:

$$E = \frac{3}{5}(0) + \frac{2}{5}(0) = 0$$

◆ Information Gain:

$$Gain = 0.971 - 0 = 0.971$$

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